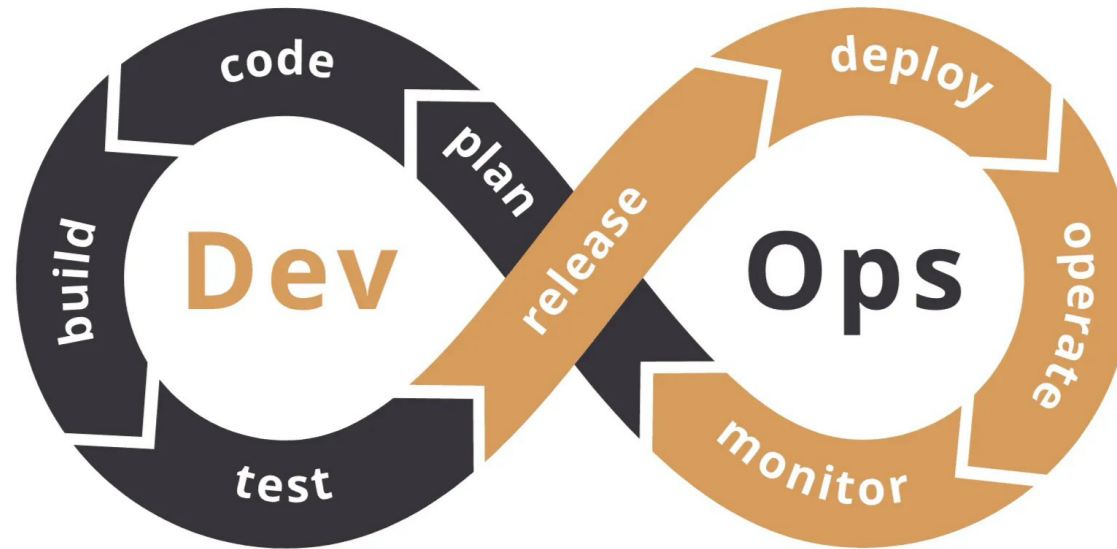




Trainer: Nilesh Ghule

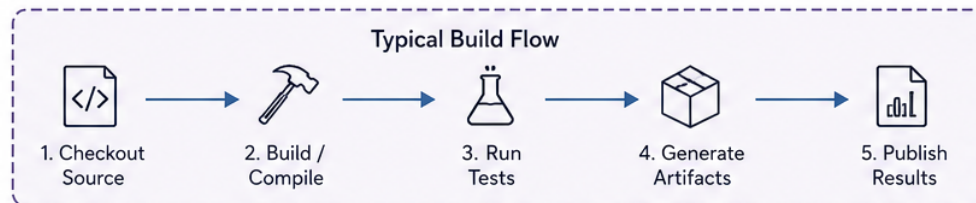
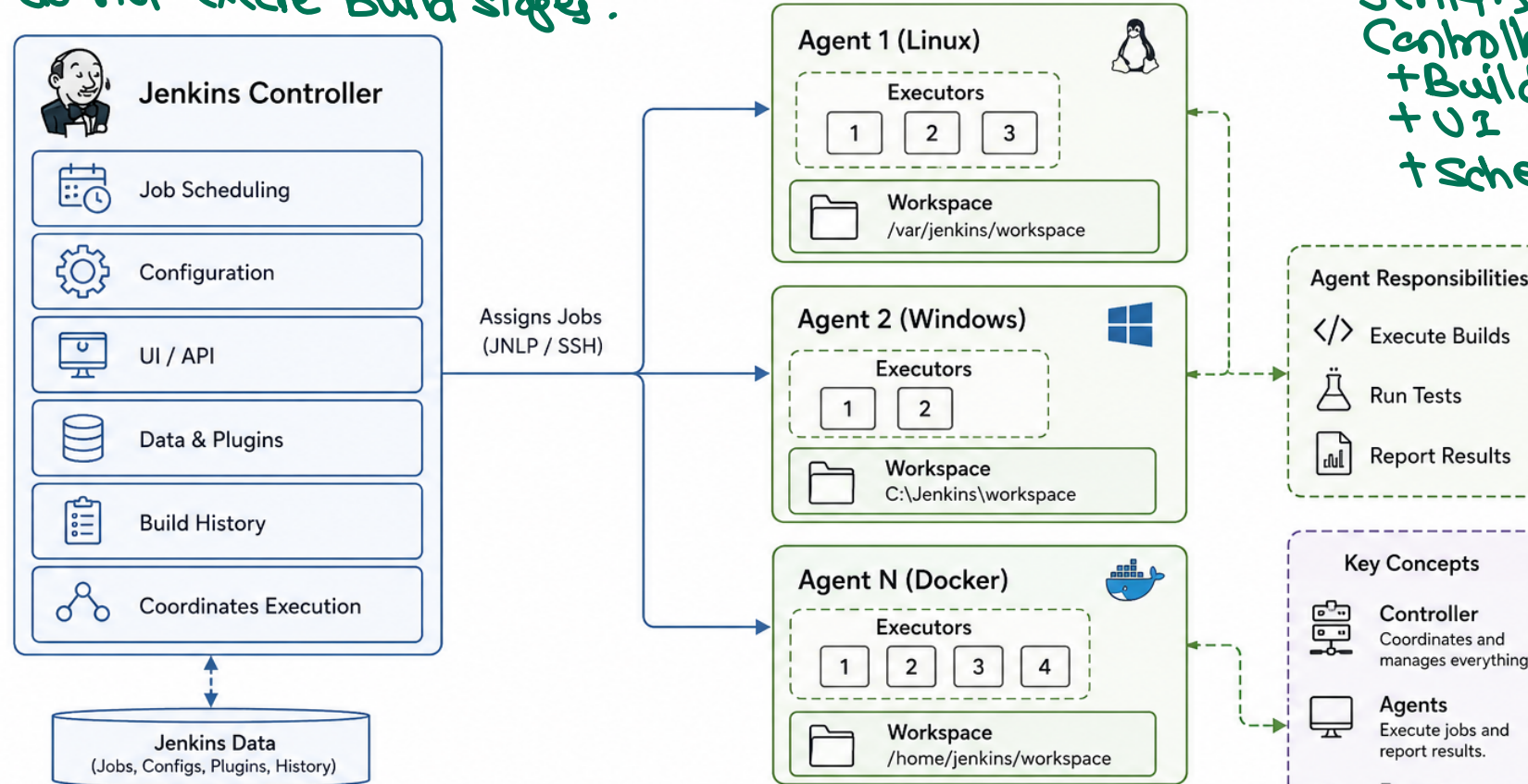
Jenkins Architecture

prod cluster
num of Executors = @
- do not execute build stages.

Jenkins Installation

option1: Single node install.
option2: Multi node install

Jenkins Architecture



on our machines.

Container Based Agent:

Jenkins Controller
+ Build
+ UI
+ Sched

pipeline {
agent {
docker {
image 'node:22'}}

stages {
stage('Build') {
steps {
sh 'npm install'
sh 'npm test'}}}

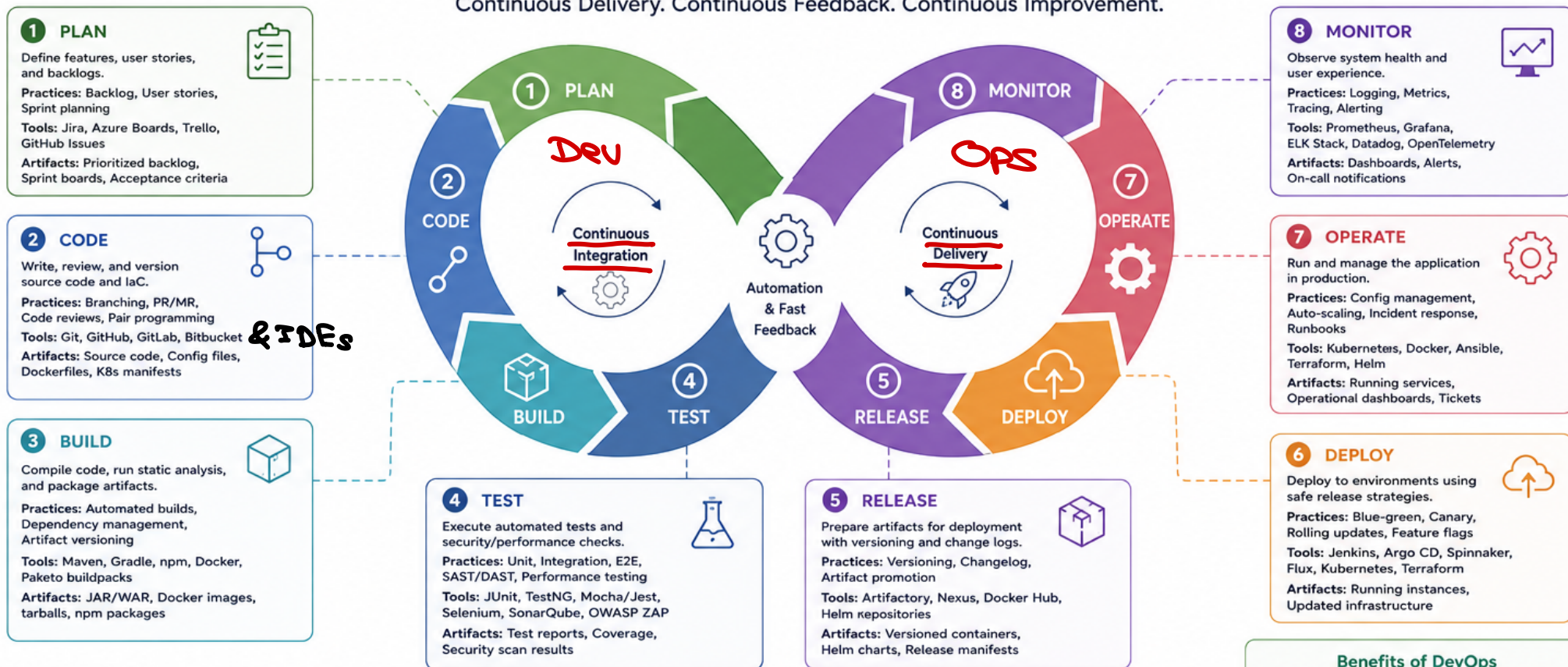
Jenkins Deploy with ssh:
sshagent(['server-key']) {

sh '''
ssh user@10.0.1.2'
docker run ...
'''

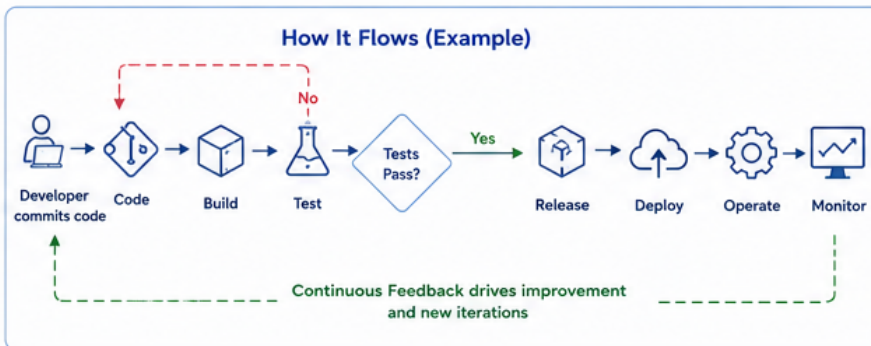


DevOps Life Cycle – The Eight Phases (∞)

Continuous Delivery. Continuous Feedback. Continuous Improvement.



DevOps Ecosystem (Key Tooling Areas)					
					
CI / CD	Source Control	Containerisation	Orchestration	Infrastructure as Code	Monitoring & Observability
Jenkins GitHub Actions GitLab CI Tekton	Git GitHub GitLab Bitbucket	Docker containerd Podman	Kubernetes Docker Swarm Nomad	Terraform Pulumi Ansible	Prometheus Grafana OpenTelemetry ELK Stack



Benefits of DevOps

- Increased Deployment Frequency**
Deploy hundreds or thousands of times per day.
- Faster Time to Market**
Features reach users sooner.
- Higher Quality**
Automated testing catches issues early.
- Greater Reliability**
Consistent environments reduce failures.
- Blame-free Culture**
Collaboration leads to higher innovation and employee satisfaction.

Key Takeaway: DevOps is a culture, a set of practices, and an ecosystem of tools that enable teams to deliver value continuously with speed, quality, and reliability.

implementing
Microservices = implementing
REST apis



Product mgmt
User mgmt
Cart mgmt
Order mgmt

monolithic
App = 1 process
↓
e.g. 4000
pools

appn all fns
split into
'independent
modules -
separate appn

Microservice arch

Product mgmt 3002
User mgmt 3001
Cart mgmt 3003
Order mgmt 3004

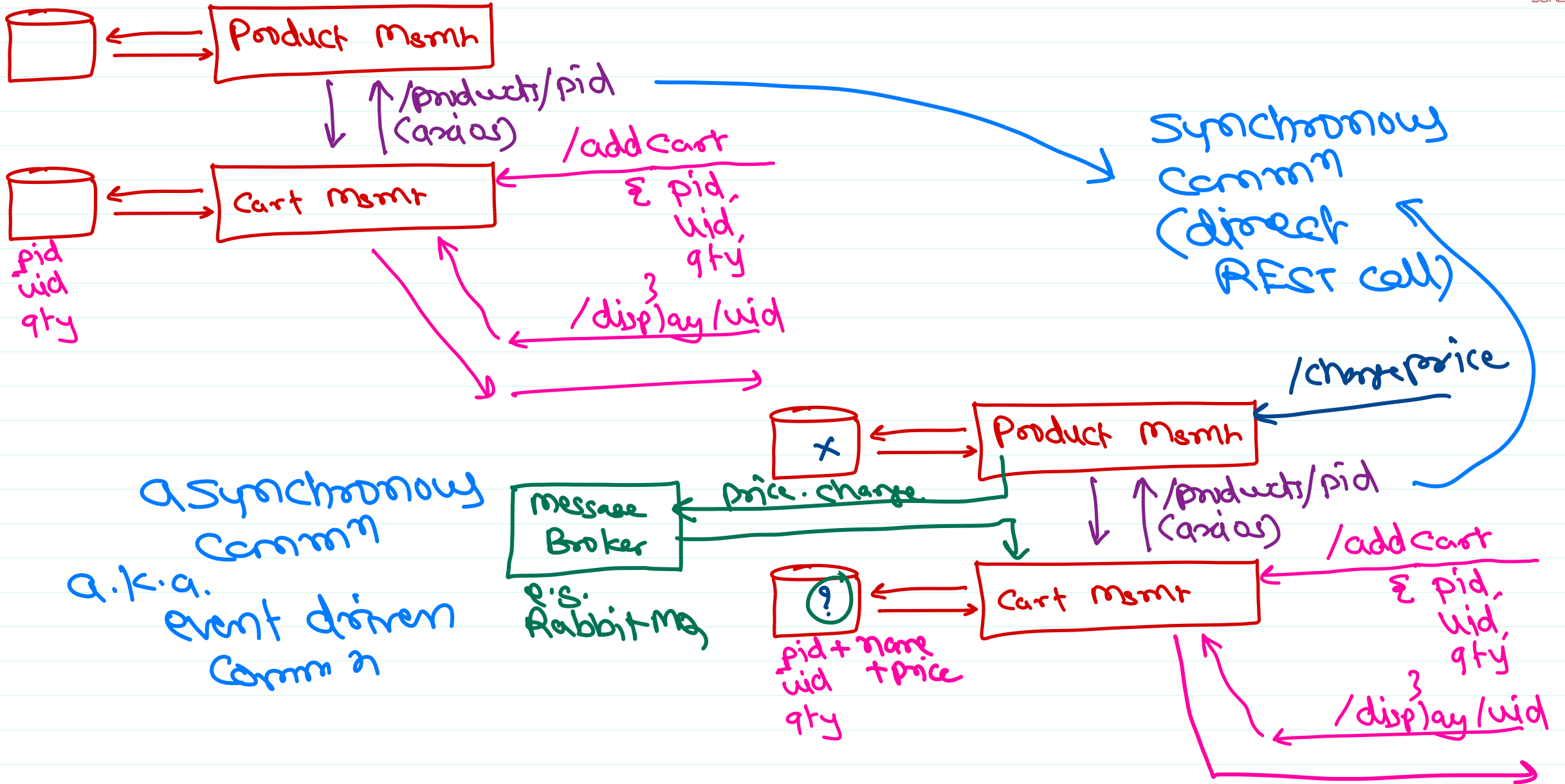
Microservice advantages

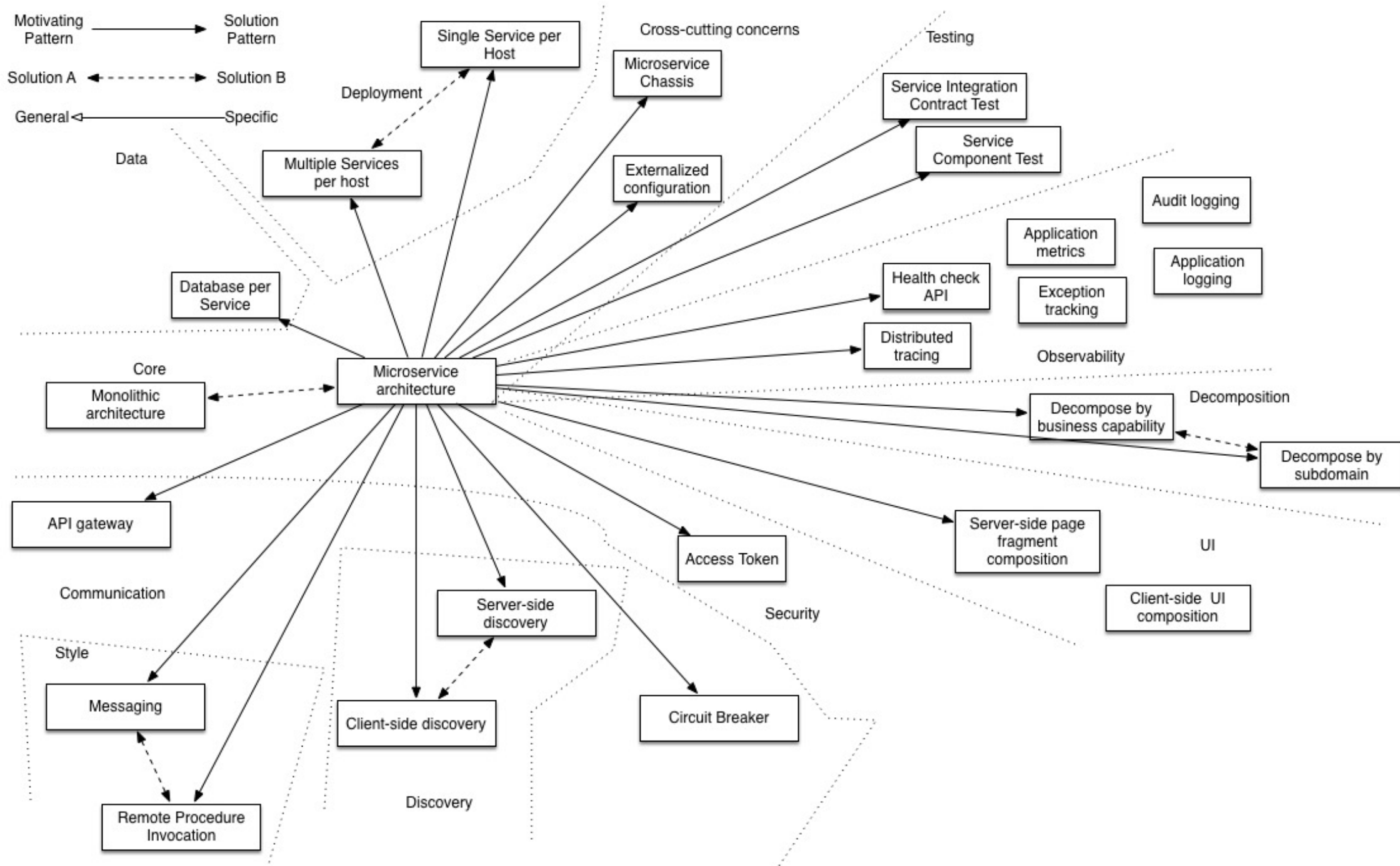
- ① small appn parts
easier to manage.
- ② small teams
- ③ tech stack may
differ
- ④ faster upgrades/
deployment
- ⑤ independent deploy
+ scaling
- ⑥ faults isolated

Microservice disadvantages

- ① app architecture &
(decomposition)
- ② app operations &
- ③ common / performance
- ④ infra ↑

Service Communication



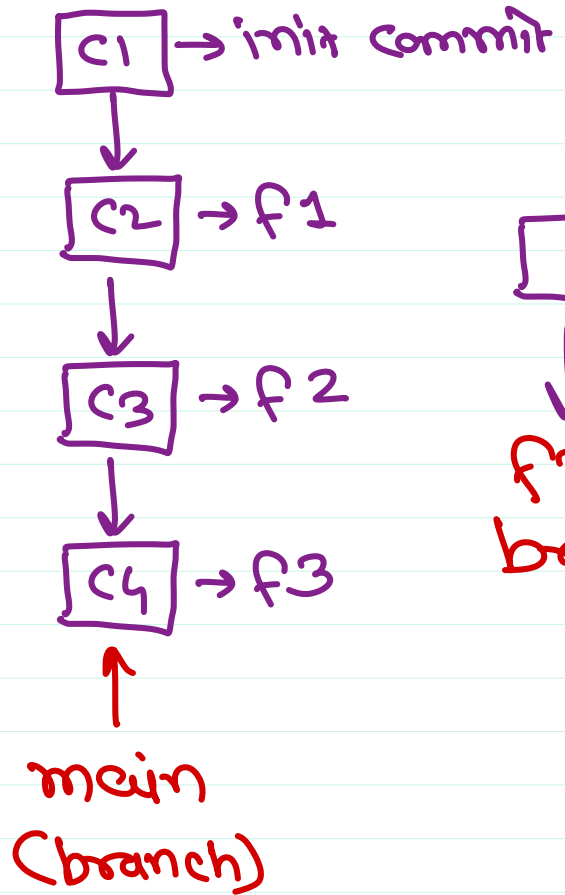


GIT



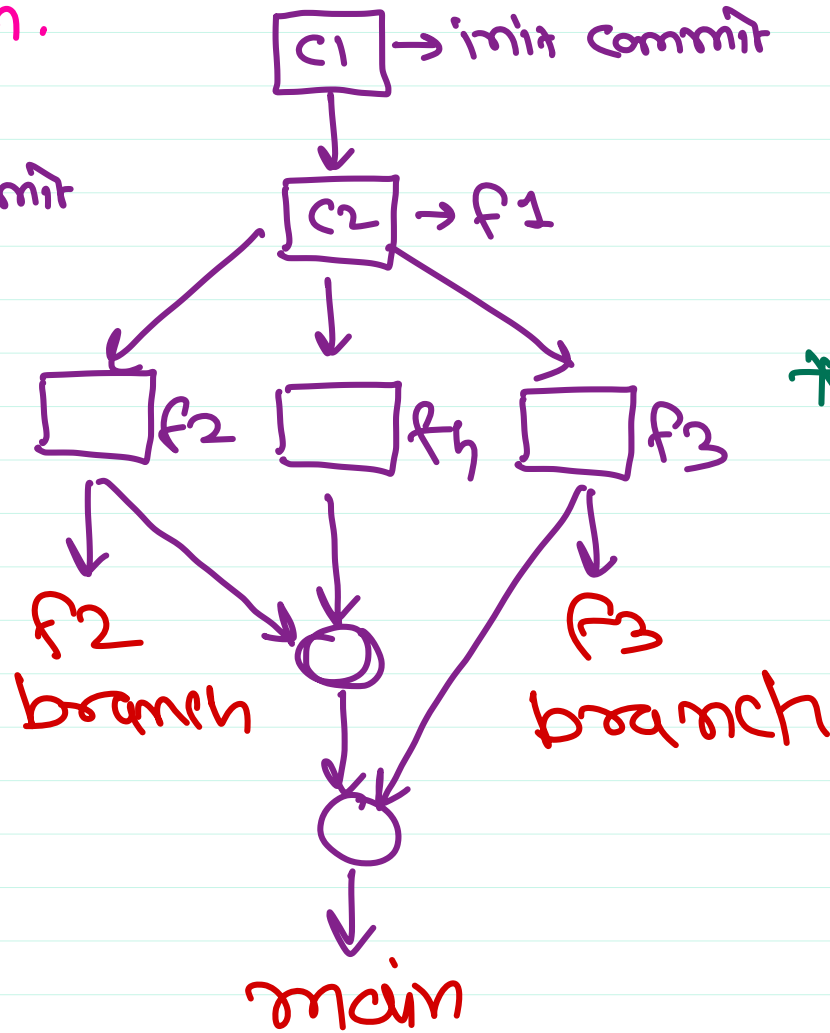
* Feature by Feature implementation.

* Naive method



parallel dev / experiments → multiple features

- ↳ by one program
- ↳ by many program



- ① git branch b1
- ② git checkout b1
- * ③ git checkout -b b2
- ④ git branch
- ⑤ git checkout main
git merge b1



Thank You!

Nilesh Ghule

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